

6.4 Pilot 14 - The living past

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The following sub-chapter presents a concise overview of Pilot 14, including its general description, targeted stakeholders and CH context, followed by the objectives and intended use of the RHDT, the pilot use case design and the key technologies and data involved.

6.4.1 Pilot 14 - Overview and Case Study Analysis

Pilot Name: The Living Past: A Digital Twin for Environmental Storytelling and Conservation

Scale: Medium

Pilot Leader: KIK-IRPA

Partners Involved: HOF, Pixelated Realities, PHASMATIC, PIN

Pilot Short Description: This pilot will assess an immersive VR/AR experience that reconstructs a monument's historical interior and exterior with its lost decoration and simulates the environmental conditions of the past. By integrating historical reconstructions with simulated IoT services, the pilot will visualise the dynamic impact of natural light and climate on both the visual appearance of the environment and the long-term degradation of its decoration. The key objective is to create a multi-layered diagnostic and storytelling tool, providing invaluable insights for conservation planning while offering heritage site and building managers unprecedented tools for building narratives with a virtual disclosure of almost inaccessible heritage sites to inform and justify restoration decisions and future preventive conservation strategies towards public authorities and the general public.

Targeted Stakeholders: The primary stakeholders are :

- Heritage Site & Building Managers will be able to disclose the full cultural and scientific story of an almost inaccessible CH asset. The VR/AR experience becomes a powerful tool to demonstrate not only the building's former splendor but also the inherent environmental vulnerabilities that necessitate its preservation. This evidence-based narrative is crucial for justifying budgets, fundraising and communicating the importance of their conservation work to stakeholders.
- Public Authorities (e.g., Ministries of Culture, National Heritage Agencies) will be able to disclose important heritage with unprecedented depth and relevance. They can offer the public an experience that shows how a building looked and why it has changed. This promotes a deeper public understanding of heritage science and the challenges of preservation, reinforcing the value of their cultural policies and investments in conservation.
- Conservators & Restorers will be able to move from observing damage to diagnosing its historical cause. By utilising the technology, they can visualise how simulated historical sunlight patterns created cumulative light exposure, or how fluctuating humidity affected its interior decorations over decades. This provides critical, evidence-based insights to inform and justify restoration decisions and future preventive conservation strategies.



While the secondary stakeholders are the following:

- Researchers & Academics can use the platform as a case study to understand the long-term effects of environmental factors, informing the design of restoration strategies and the impact of climate change.
- The General Public will receive a far more engaging and educational experience than a simple visual reconstruction, learning about the interplay of art, science and time.

Involved Stakeholders: No external stakeholders are defined for Pilot 14.

Historic Background of CH Asset: The Japanese Tower in Laken (Brussels) is a well-known landmark for Brussels and Belgium. It was commissioned by King Leopold II and officially opened in 1904. The Tower looks very Japanese in appearance, but by no means it is an oriental object placed in a distant world: it is a monumental expression of how Japanese art overwhelmed Europe during the Fin de Siècle, a rare place where the best aesthetics of two worlds merge.

This five-floored royal jewel was lavishly decorated with a variety of mixed materials, including polychromies, metal, paint, wallpaper, stained glass, linoleum, textile and - most prominently - exquisite Japanese lacquer panels, one of Japan's most distinctive artistic art forms. The lacquers of the Tower represent a broad selection of provenances, styles and colours. After more than 100 years, most floors are still conserved untouched, but their conservation condition is very worrisome.

Crucially for this pilot, the Japanese Tower possesses an extraordinary degree of authenticity and integrity. Structurally, the building remains largely unaltered since its construction and even its original interior heating system has not been changed. This means its interior climate is almost entirely governed by its natural relationship with the outdoor environment. This unique lack of modern intervention makes the Japanese Tower an invaluable "time capsule". The historic interior and exterior have undergone a century of natural ageing, offering an unparalleled opportunity to study the long-term effects of environmental conditions on its specific materials and decorations. While currently closed to the public due to accessibility issues, it remains a beloved and iconic landmark, its story deeply intertwined with the legacy of king Leopold II and the cultural ambitions of his time.

Conservation status of CH Asset: The Japanese Tower's primary conservation challenge lies not in its structural integrity, but in the complex and fragile state of its unique interior and exterior decorations. While recent works have resolved issues with the roof's water tightness, the building's original materials bear the marks of over a century of natural ageing. The treatment of these decorations is exceptionally complex because they represent a rare fusion of Japanese and European fin-de-siècle artistic techniques and materials, a combination that has not been thoroughly studied from a conservation-restoration perspective.

A significant issue is the fading of the original vibrant colours due to cumulative light exposure. To mitigate further weather-induced damage, the exterior decorative panels have been carefully dismantled and are currently stored inside the tower's first floor, while the interior panels remain in situ. This situation presents a unique opportunity: certain 'historic' panels are decorated on both sides, with one side exposed to the interior and the other



protected from light. This provides a direct, physical reference for understanding the original decorative finishes and quantifying their degradation.

Consequently, this pilot is essential because standard conservation cannot begin without a thorough preliminary study. The digital twin approach will serve this purpose, allowing for the virtual reconstruction of the Tower's original splendor while also providing a platform to study the degradation caused by light, temperature and humidity, thus informing future physical conservation-restoration strategies.

Regulatory and legal frameworks: The Japanese Tower is a protected monument under the legislation of the Brussels-Capital Region and is managed by the Belgian Buildings Agency (Regie der Gebouwen/Régie des Bâtiments), the property manager of the Belgian Federal State. The Japanese Tower is property of the Belgian Federal State and considered also part of the collection of the Royal Museum of Art and History (a Federal Scientific Institute). Any intervention, including research and monitoring activities, falls under the purview of these authorities. In addition, some historic furniture elements are stored in their collection.

The pilot's implementation, while primarily digital, requires a formal collaboration agreement with the Belgian Buildings Agency and any relevant heritage bodies. This agreement, rather than a building permit for physical alteration, will specify the scope of the non-invasive activities, such as 3D laser scanning and the temporary placement of IoT sensors. As the Royal Institute for Cultural Heritage, a Federal Scientific Institute as well, has already been preparing a research project since a few years to study the conservation-restoration of the lacquers of the Japanese Tower, we have already established the necessary agreements to have access to the Tower for, amongst others, the digital reconstruction of its interior and exterior. Henceforth, there is no regulatory or legal restriction to include the Japanese Tower as a pilot case.

Restrictions and limitations: The pilot's implementation is primarily digital and non-invasive, focusing on the aggregation and annotation of existing data, its digital reconstruction and the installation of IoT sensors. Therefore, it does not require specific permits for physical intervention.

The collaboration has, however, yet to detail all the restrictions and rights related to the data and their intellectual property rights. This includes defining the ownership of the 3D models and raw data, establishing protocols for data sharing among partners and setting clear guidelines for the publication or public dissemination of findings to protect the asset's security and the integrity of the research. As the involved parties are public federal authorities, these protocols will be developed in full alignment with the European directive on open data. This provides a strong framework that anticipates a straightforward process for the publication and public dissemination of the project's findings.

6.4.2 Pilot 14 - RHDT Objectives

This section defines the specific aims and uses of the pilot RDHT, including both high-level goals and specific tasks.

Primary Goals:

- **Create a scientifically-backed, 4-dimensional reconstruction** to develop a virtual laboratory environment where users can navigate through time and explore the pilot's documented historical appearance, including its original vibrant colour schemes and now-dismantled exterior decorative elements.
- **Simulate and visualise the historical environmental impact** to develop and integrate a simulated historical climate model within the digital twin. This model will use the building's known properties and historical weather data to visualise the cumulative impact of natural light, temperature and humidity on specific materials, showing why and how fading and degradation occurred.
- **Establish a structured data repository for future research** to create a rich, structured dataset correlating environmental factors, visitor presence and material decay over time and where researchers, heritage building managers and conservator-restorers can access and visualise diverse datasets without needing specialised technical expertise. This repository will serve as a valuable resource for researchers studying the long-term impacts of climate change on CH.
- **Facilitate informed, data-driven decision-making** to empower heritage managers to perform simplified risk analysis and run basic forecasts based on integrated data. The goal is to provide them with the evidence needed to justify conservation actions, optimise maintenance schedules and manage visitor impact more effectively. Implement functionalities that allow users to easily navigate and visualise the historical evolution of environmental conditions, enabling them to correlate past events (e.g., a heatwave) with physical changes or damage to the asset.
- **Generate quantitative data to inform future conservation strategies** to provide conservator-restorers with concrete data on historical light exposure and climate impact. This data can be used to model and compare the long-term effects of different potential restoration treatments or environmental control strategies, creating a robust, evidence-based foundation for the future physical restoration.

Secondary Goals:

- **Establish an interdisciplinary educational platform** to serve as a rich case study for universities and research institutions, allowing students and academics to explore the complex intersection of architectural history, material science, conservation ethics and digital reconstruction techniques within a single, interactive environment.
- **Deliver a high-fidelity, immersive VR/AR experience for public disclosure** to create a compelling, publicly accessible virtual experience that grants access to the inaccessible monument. The goal is to allow the public to explore its reconstructed splendor and understand the powerful story of its evolution and the scientific challenges of its preservation.



- **Enable proactive condition monitoring and alerting** to move heritage management from a reactive to a proactive model by implementing a system that automatically analyses data streams, detects when conditions will exceed safe thresholds and issues clear, timely alerts to managers about potential risks.
- **Establish the foundation for automated environmental control** to demonstrate that the RHDT can lay the technological groundwork to show how the system can be used to automatically steer environmental controls (e.g., activating (de)humidifiers, adjusting lighting, etc.) to maintain optimal preservation conditions with minimal human intervention. Demonstrate the RHDT's capacity to simulate the impact of environmental control actions.

Technological Demonstrations:

- **RHDT Platform:** The pilot will demonstrate an integrated application environment that acts as the main pilot environment. It will showcase how a heritage digital twin can be 'reactive' in response to new data.
- **Advanced Environmental Simulation Engine:** The pilot will demonstrate an advanced environmental simulation engine capable of modelling historical solar paths and climate conditions to calculate and visualise their long-term impact on specific, accurately mapped materials within the 3D model. This engine will also be used to model future climate scenarios and environmental control actions.
- **Semantic Linking and Data Integration Engine:** The pilot will showcase the backend technology used to connect unstructured data (like a paragraph in a PDF report) or structured database entries to specific geometric locations or components on the 3D digital twin.
- **Seamless integration of IoT data streams:** The pilot will demonstrate the real-time ingestion and processing of data from diverse IoT sensors monitoring critical parameters such as temperature, relative humidity (RH), light levels, etc.
- **Spatiotemporal data fusion and visualisation:** The pilot will showcase the fusion of time-series sensor data with 3D spatial models (3D/AR/VR). This will allow users to visualise not only what is happening (e.g., humidity rising) but exactly where in the building it is occurring, mapped onto a 3D representation of the asset.
- **Interactive 2D/3D visualisation engine:** The pilot will demonstrate the use of a high-performance graphics engine to render the heritage asset and dynamically overlay complex datasets, making them visually accessible and easy to interpret on standard computer hardware.
- **Predictive analytics and risk simulation:** The pilot will demonstrate the application of predictive algorithms that use historical and real-time data to forecast future conditions and simulate potential risks. This will be shown through intuitive dashboards that display risk levels (e.g., as colour-coded heatmaps) and generate alerts.

6.4.3 Pilot 14 - Use Case Design

A comprehensive design for the pilot is presented here, including user interactions and timelines.

The pilot use case for the Japanese Tower will be developed by creating a sophisticated RHDT centered on a powerful, data-driven predictive model. The foundational step involves the installation of a comprehensive IoT sensor network to capture a broad range of indoor and outdoor climate data for at least one full year, supplementing the limited historical data already available. The primary objective of this extensive data collection campaign is to train a highly-tuned Predictive Indoor Climate AI model that learns the unique and authentic hygrothermal behaviour of the tower, which has remained largely unaltered since its construction. Once this building-specific model is validated, it will be used for two distinct but complementary applications. Firstly, it will enable the reconstruction of the past by feeding it with historical outdoor climate data (available, with varying levels of detail, back to 1905) to simulate the long-term indoor environmental conditions that led to the degradation observed today. Secondly, the model will enable the simulation of the future by driving it with future climate projection data, allowing stakeholders to forecast long-term risks and plan for proactive conservation. All of this simulated data, both historical and future, will be integrated and visualised on the high-fidelity 3D models of the tower within an immersive VR/AR environment, creating a powerful tool for environmental storytelling, diagnosis and strategic planning. To demonstrate the practical value of this powerful diagnostic and predictive system for different professional needs, the pilot will test a series of user-centered scenarios (further highlighting the development and objectives of the pilot case):

- **Test Scenario 1: Diagnosing Differential Fading**
 - Objective: To use the RHDT as a diagnostic tool to provide quantitative evidence explaining why similar panels have degraded differently.
 - User Role: Conservator-Restorer or Conservation Scientist
 - Scenario Steps:
 - The conservator observes that an interior panel on the south-facing wall of the third floor is significantly more faded than a panel of the same material and age on the north-facing wall. The hypothesis is that cumulative solar exposure is the cause.
 - Using the VR version of the RHDT, the conservator virtually stands in the reconstructed 1905 hall. They select the south-facing panel and activate the "Environmental History" function.
 - The system runs the simulation and displays a data card: "Total Estimated Light Exposure (1905-2024): 48.5 million Lux hours." A heatmap overlay visually confirms the most exposed areas.
 - The conservator then navigates to the north-facing panel and repeats the process. The system returns a value of "Total Estimated Light Exposure (1905-2024): 4.2 million Lux hours."
 - The tenfold difference in quantitative light dose provides definitive evidence that solar radiation is the primary driver of the differential fading. This conclusion is logged as a permanent annotation,

informing the conservation strategy for both panels. The quantitative data provided is clear, comparable and directly supports the conservator's expert hypothesis.

- **Test Scenario 2: Justifying a Restoration Budget**

- Objective: To use the RHDT as a powerful communication and justification tool to secure funding from a non-expert governing body.
- User Role: Heritage Site & Building Manager
- Scenario Steps:
 - The Building Manager presents a multi-million euro restoration proposal to a committee of public officials. Instead of a PowerPoint, they use the VR/AR experience.
 - The "Wow" Factor: They first immerse the committee in the vibrant, fully reconstructed 1905 interior, establishing the asset's immense cultural value and what has been lost.
 - The "Why" Factor: The manager then activates the environmental simulation layer. The committee watches an accelerated time-lapse showing decades of sunbeams sweeping across the room, visibly and dramatically fading the brilliant colours of the virtual decorations.
 - Finally, the manager shows the "as-is" 2024 state, with the dismantled exterior panels stored inside, visually demonstrating the urgent need for intervention that the simulation explained.
 - The immersive, evidence-based narrative transforms an abstract budget request into a tangible, urgent and understandable mission to save a national treasure. Post-experience feedback from a test audience confirms the experience is more persuasive and informative than a traditional presentation.

- **Test Scenario 3: Virtual Prototyping of Environmental Controls**

- Objective: To use the RHDT as a predictive tool to design and test the effectiveness of potential environmental interventions before any physical implementation.
- User Role: Conservation Engineer or Specialist Architect.
- Scenario Steps:
 - An engineering office is tasked with designing a window shading solution for the south-facing windows that reduces harmful light exposure without compromising the historic character.
 - The engineer first runs the baseline solar simulation in the RHDT, confirming the high light dose on the interior panels.
 - Using the RHDT's "Intervention Modelling" tool, they virtually add "Option A: a 99% UV-filtering film" to the windows and re-run the simulation. The system shows UV is blocked, but the total visible light dose is only reduced by 15%.
 - They then model "Option B: historically appropriate exterior canvas awnings." They re-run the simulation. The system shows this option reduces the total light dose by 60% and creates a diffuse, less damaging light.



- By comparing the quantitative results of both virtual prototypes, the engineering office can confidently recommend the awnings as the most effective solution, providing a report with clear data visualisations to justify their choice.

Target Audience/Users: The following outlines the primary user groups of the pilot, along with their roles, profiles and modes of interaction.

- Heritage Site Managers and Public Authorities are considered as the primary users. Their technical profile is typically non-specialist, with expertise in heritage management, policy and public communication, but not in 3D-modelling or environmental simulation. They will primarily interact with a polished, presentation-focused version of the RHDT on desktops, tablets, or large screens. Their main goal is to use the tool for storytelling and justification. They will switch between the reconstructed "as-built" state and the current "as-is" state and activate the pre-set environmental simulation animations to visually communicate the asset's value, its historical degradation and the urgent need for conservation funding to stakeholders and the public. They will also guide VIPs or committees through the AR/VR experience in a controlled setting, as such these AR/VR experiences can be used as guided storytelling events.
- Conservator-Restorers, Heritage Scientists and Researchers are considered as the core expert users. These users have a specialised technical background in material science, conservation, art history, or architecture. They are experts in data analysis within their domain but require an intuitive interface to access the pilot's complex simulations. They will primarily interact with a feature-rich version of the RHDT on a desktop computer for in-depth analysis and use VR for immersive review. They will use the system to run environmental simulations, query quantitative data on light exposure, compare different chronological states of the asset and create detailed annotations to build an evidence-based conservation strategy.
- Engineers and Conservation Architects are technical experts in fields like lighting design or environmental engineering, comfortable with CAD software and data-driven design. They will use the RHDT on a desktop computer as a "virtual sandbox." They will interact with a specific module that allows them to virtually place new elements (e.g., 3D models of UV films or window shadings) onto the building model. They will then re-run the environmental simulation to generate comparative data that proves the effectiveness of their proposed interventions before any physical implementation.
- The General Public as the end users consist of novices with no technical background. The experience must be immediately engaging and self-explanatory. Their interaction will be through simplified, robust platforms. This could be a VR headset in a dedicated museum kiosk offering an immersive tour, an on-site AR application on their mobile device showing the reconstructed decorations overlaid on the real building, or a Web-based 3D viewer for remote exploration. Their interaction is focused on guided discovery and narrative.

Pilot Scope: The pilot will focus on the **complete digital reconstruction and environmental simulation** of the Japanese Tower. It will create and visualise two primary

historical states: the "as-built" splendor of c. 1905, with its original vibrant colour schemes and fully installed decorative panels and the current "as-is" condition.

The environmental simulation will specifically model historical solar radiation to calculate the cumulative light dose on surfaces, as well as general long-term temperature and humidity trends based on historical climate data. The project's final deliverables will include two distinct applications: **an expert-level analytical tool for diagnostics and intervention planning and a polished, narrative-driven VR/AR experience for public disclosure.**

To maintain a focused and achievable scope, the pilot will not reconstruct a continuous, year-by-year evolution of the Tower. The environmental simulation will not involve complex Computational Fluid Dynamics (CFD) for air movement or micro-particulate simulation.

Pilot Timeline: The following outlines the phased timeline of the pilot, detailing the main activities and their temporal progression.

- **Phase 1: Data Collection, Data Acquisition & Model Acquisition** (months 7-20)
 - T.1.1. Data collection (months 7-14)(P14.T1). In-depth archival research to source historical plans, photographs and material descriptions. Most of the materials are already available.
 - T.1.2. Setup and calibration of data acquisition (months 10-14)(P14.T2). This involves the on-site installation, network configuration and calibration of all IoT sensors (temperature, RH, etc.) to establish a reliable, real-time data stream.
 - T.1.3. 3D model development (months 13-20)(P14.T3). In parallel, this activity covers the high-resolution 3D documentation (e.g., laser scanning and photogrammetry) of the CH asset and the processing of this data to create the foundational geometric model for the RHDT.
- **Phase 2: Core Platform and Model Development** (months 13-28)
 - T.2.1. RHDT development and integration (months 19-26)(P14.T4). This is the primary software development phase. It includes building the core RHDT platform, integrating the live sensor data feeds and developing the main user interface, visualisation engine and primary alerting system.
 - T.2.2. Development of simulation models (months 13-28)(P14.T5) In parallel development of advanced simulation models, such as a lighting/solar radiation model and an indoor climate prediction model.
- **Phase 3: Advanced Integration and System Finalisation** (months 25-30)
 - T.3.1. Integration of simulation models in the RHDT (months 25-30)(P14.T6). The specialised simulation models (lighting) are integrated into the main RHDT platform, allowing their outputs to be visualised and correlated with sensor data. The full system is finalised and prepared for user testing.
- **Phase 4: User Engagement and Validation** (months 27-32)
 - T.4.1. Testing and validation with stakeholders (months 27-32) (P14.T7) The completed RHDT is deployed for use by the target stakeholders (heritage managers, curators). This phase includes user training, running predefined scenarios, gathering feedback and validating the system against the success criteria.



- **Phase 5: Evaluation and Reporting** (months 32-34)
 - T.5.1. Final analysis, evaluation and reporting (months 32-34) (P14.T8) All quantitative data and qualitative feedback from the validation phase are analysed. The final pilot report is compiled, detailing the outcomes, achievements and recommendations for future exploitation.

	WP7/9/11												WP7/9/11						WP8/10/12						WP8/10/12												
	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	
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T.4.1																										✓	✓	✓	✓	✓	✓	✓					
T.5.1																																✓	✓	✓			

Table 6.16 - Pilot 14 - Implementation Gantt Chart

Key Activities:

- High-resolution 3D documentation (laser scanning/photogrammetry) of the pilot case
- Archival research into historical plans, photographs, inventories and material studies to gather the evidence required for the historical reconstruction
- Detailed condition monitoring and photographic documentation of the pilot case
- Installation and calibration of IoT sensors (temperature, relative humidity, light, dust particles, carbon dioxide) in key locations
- Development of two distinct, high-fidelity 3D models: the current "as-is" state and the historically accurate, fully decorated "as-built" state
- Development of the Web-based RHDT platform
- Integration of the real-time IoT data streams
- Development and integration of the lighting simulation
- Development and integration of the indoor climate prediction model
- Development and integration of the risk assessment tools
- Model calibration using the real-time IoT data to validate and calibrate the environmental simulation model
- Implementation of the interactive 3D visualisation engine and the temporal data analysis tools
- Conducting training workshops for heritage managers and conservators
- Running the defined test scenarios and gathering qualitative feedback through user interviews

- System optimisation based on user feedback and performance metrics

Success Criteria: The success of the pilot will be assessed based on the following quantitative and qualitative criteria.

- **Quantitative Metrics:** *i)* Simulation-to-Reality Correlation: The primary diagnostic goal is considered met if the simulated environmental data shows a strong correlation with observed reality with threshold: The cumulative light exposure heatmap generated by the simulation must show a greater than 80% visual correlation with the known fading patterns on a selected control panel (e.g., the two-sided panels), *ii)* IoT Data Reliability: The real-time sensor network used for model calibration must be stable and reliable with threshold: The IoT data pipeline must maintain greater than 99% uptime during the 12-month active data collection and calibration phase, *iii)* VR/AR Application Performance: The immersive experience must be comfortable and seamless to prevent user discomfort and maintain engagement with threshold: The VR application must consistently maintain a stable frame rate of greater than 72 Frames Per Second (FPS) on the target hardware during all tested scenarios, *iv)* Expert User Task Completion Rate: The analytical tools must be intuitive enough for experts to use effectively after training with threshold: After a single training session, more than 90% of trained expert users (conservators, engineers) must be able to complete their designated test scenario without requiring assistance.
- **Qualitative Metrics:** *i)* Diagnostic and Planning Value: The tool must be perceived as a valuable asset for conservation and engineering professionals with threshold: In a post-pilot survey, expert users must give an average score of at least 4 out of 5 to the question: "How valuable is this tool for diagnosing historical damage and planning future interventions?", *ii)* Storytelling and Persuasiveness: The VR/AR experience must be an effective communication tool for justifying conservation efforts with threshold: Greater than 80% of test users from the managerial/public authority group must "agree" or "strongly agree" that the immersive experience is a more persuasive tool for justifying budgets than traditional methods like presentations and reports *iii)* Public Engagement and Knowledge Transfer: The public-facing experience must be both engaging and educational with threshold: A post-experience survey must show that greater than 85% of public users found the experience "engaging" or "highly engaging," and can correctly answer at least two out of three basic questions about the Tower's history and conservation challenges, demonstrating successful knowledge transfer.

6.4.4 Pilot 14 - Technologies and Tools Involved

Pilot 14 will utilise the technologies listed below:

- Phasma3D Viewer (CAW-01)
- Navigational Hypertext Service (BEDP-02)
- Annotation Service (BEDP-03)
- Hypertext Knowledge Base Service (BEDP-05)
- IoCT Services (BEDP-07)
- Predictive Indoor Climate AI Service (BESS-07)



- Risk Assessment Service (BESS-08)
- Illumination Simulation Services (BESS-13)
- VR collaborative environment (CAVR-07)

Detailed descriptions of these technologies are provided in Chapter 8 - Software Stack.

6.4.5 Pilot 14 - Available Data Resources and Integration

The following section presents the existing datasets and data resources to be utilised in the pilot, including their integration and enhancement approaches.

Available Data Types: As this is a data-centric pilot, a rich variety of existing documentation will be leveraged. Data will be documented as it is gathered for the pilot cases. It covers: *i)* Heritage documentation data: An extensive inventory of reports, scholarly articles and historical records is available. These are primarily in.pdf and.docx formats, *ii)* Photographical documentation data: Historical black-and-white photos, RGB or multispectral digital photos of the Japanese Tower (.jpg,.tiff), *iii)* Analytical data, including: technical imaging such as high-resolution scientific imagery and analytical data such as images of paint cross-sections, graphs from data analysis, etc.

Data Integration Requirements: The following outlines a set of indicative fields comprising the core information to be formalised and incorporated into the ARTEMIS KB.

KB Integration: The ARTEMIS KB will serve as the central metadata registry for all data sources, simulation models and analytical outputs generated within Pilot 14. The primary goal is to ensure that all information is Findable, Accessible, Interoperable and Reusable (FAIR). Instead of storing the raw, high-volume time-series data or large 3D model files itself, the KB will store structured metadata records. Each record describes a specific data asset (like a sensor) or an analytical event (like a simulation run) and provides a persistent link (URL or API endpoint) to where the raw data can be accessed from its host datastore (e.g., the KIK-IRPA server). A unified metadata schema, as detailed below, will be applied.

- **IoT Service Integration: Registering Physical Sensors:** Each physical sensor deployed as part of the IoT network at any of the sites will be registered as a unique asset in the KB. This creates a comprehensive, queryable inventory of the entire monitoring infrastructure.

Field Title	Type	Description	Temporary Example
NodeID	STRING	Unique identifier for the physical IoT node	louvre- <i>nap3</i> -node-sw01
SensorID	STRING	Unique identifier for a specific sensor on the node	louvre- <i>nap3</i> -node-sw01-scd30-co2
LocationDesc	STRING	Human-readable description of the sensor's placement	Louvre, Napoleon III Apts, Grand Salon, SW Facade Wall
SensorType	STRING	The model and type of the sensor	Sensirion SCD30 CO2 Sensor



DataTypes	STRING	The physical quantities being measured (comma-separated)	co2_concentration, temperature, relative_humidity
DataUnits	STRING	The units for each corresponding data type (comma-separated)	ppm, celsius, percent_rh
DeploymentDate	DATETIME	ISO-8601 date and time when the sensor became active	2025-04-10T11:30:00 Z
Status	STRING	Current operational status of the sensor (e.g., Active, Inactive, Maintenance)	Active
LastMeasurement	DATETIME	ISO-8601 timestamp of the last successfully registered measurement	2025-10-24T23:15:00 Z
Interval	STRING	The intended time interval between measurements (e.g., ISO 8601 duration format)	PT15M (15 minutes)
DatasetURL	URL	Persistent URL/API endpoint to access the time-series data	https://artemis.kikirpa.be/data/ts/louvre-nap3-node-sw01-scd30

Table 6.17 - Pilot 14 - Knowledge Base Integration - IoCT Service

- **Predictive AI Service Integration: Registering Forecasts:** For each forecast generated by the Predictive Indoor Climate AI Service, a record will be created in the KB. This allows the RHDT to retrieve, display and act upon future predicted conditions

Field Title	Type	Description	Temporary Example
ForecastID	STRING	Unique ID for the specific forecast or simulation run	blenheim-lib-forecast-72h-20251201
AssetID	STRING	Unique ID of the CH asset / RHDT context being forecasted	rhdt:woodstock-blenheim-library
ModelID	STRING	Identifier of the trained AI model used for the forecast	BlenheimLib-Hygrothermal-v1.3-LSTM
ForecastHorizon	STRING	The time period the forecast or simulation covers	72 hours 2050-01-01 to 2051-01-01
InputClimateSource	JSON	Specifies the source of the external climate data used as input	{"type": "api", "endpoint": "https://api.open-meteo.com/..."}
InputDataURL	URL	URL to the exact archived data (e.g., JSON or CSV) retrieved from the source and used as input for this forecast	https://artemis.archive.kikirpa.be/inputs/fc-2025.../weather.json
ForecastOutputURL	URL	URL/API endpoint to access the archived forecasted time-series data	.../data/forecast/blenheim-lib-72h-20251201
Curator	STRING	Person/system that validated the record	ai.ops@kikirpa.be



ForecastTimestamp	DATETIME	ISO-8601 timestamp of the forecast run	2025-10-24T23:15:00Z
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Table 6.18 - Pilot 14 - Knowledge Base Integration - Predictive AI Service

- **Risk Assessment Service Integration: Registering Risk Events:** Each time the Risk Assessment Service identifies a current or predicted risk that breaches a user-defined threshold, it will generate a "Risk Event" record in the KB. This provides an actionable and auditable log of all significant threats to the collection.

Field Title	Type	Description	Temporary Example
EventID	STRING	Unique identifier for the specific risk event	evt-gruuthuse-20251115-mold-01
AssetID	STRING	Unique ID of the CH asset / RHDT context affected	rhdt:bruges-gruuthusemuseum
LocationDesc	STRING	Human-readable location of the risk event	Gruuthusemuseum, Second Floor, Woodworking Gallery
RiskType	STRING	The type of conservation risk identified	Mold Growth Risk (Predicted)
RiskLevel	STRING	The severity of the risk (e.g., Low, Medium, High, Critical)	High
RiskBasis	STRING	Indicates if the risk is based on Real-time data, a short-term Forecast, or a long-term Climate Projection	Forecast
RiskMetrics	JSON	Quantitative data detailing the risk, including the value that triggered the alert, the threshold breached and duration if applicable	<pre>{ "predicted_rh_percent": 78.5, "threshold_rh_percent": 70.0, "forecasted_duration_hours": 12 }</pre>
EventTimestamp	DATETIME	ISO-8601 timestamp for when the risk is predicted to occur or was detected	2025-11-17T18:00:00Z
CreationTimestamp	DATETIME	ISO-8601 timestamp for when the risk event was logged in the KB.	2025-11-15T18:05:00Z
TriggeringSource	STRING	The RunID, ForecastID, or SensorID that provided the data causing the alert	ForecastID: gruuthuse-forecast-72h-20251115
NotificationLog	JSON	An auditable log of all notifications sent for this event. Each entry records the destination (target), the time it was sent (timestamp) and its delivery or response status	<pre>[{ "target": "heritage-alerts@kikirpa.be", "timestamp": "2025-11-15T18:05:10Z", "status": "sent" }, { "target": "sms:+32475123456", "timestamp": "2025-11-15T19:05:00Z", "status": "acknowledged" }]</pre>

Table 6.19 - Pilot 14 - Knowledge Base Integration - Risk Assessment Service



Data Format Compatibility: The RHDT platform must integrate a diverse array of data types from multiple sources. To ensure seamless interoperability between data acquisition, the KIK-IRPA simulation services, the ARTEMIS KB and the front-end visualisation tools (3D Viewer and AR/VR collaborative workspace), a clear and standardised data format strategy is required.

- **3D Spatial Data Formats:** The 3D models are the foundational canvas of the RHDT. A multi-format approach will be used to serve different purposes:
 - For Immersive VR/AR: All 3D meshes intended for the collaborative VR/AR environment will be delivered as glTF Binary (.glb). These assets will be rigorously optimised to meet the constraints of the Meta Quest 3's mobile XR platform. This includes: Polygon Budgets: Reducing mesh complexity to ensure high frame rates, Texture Optimisation: Using compressed texture formats (e.g., KTX2) to minimise memory usage, Material and Draw-Call Limits: Simplifying materials and mesh structures to ensure efficient rendering. The goal is to maintain a smooth, interactive experience (targeting ≥ 72 FPS) within the AR/VR collaborative workspace.
 - For Simulation Services: The KIK-IRPA simulation services require clean, watertight 3D mesh geometry as input. These models will be provided in standard formats such as .obj or .glb.
- **Environmental and Simulation Data Formats:** To ensure that real-time, forecasted and simulated environmental data can be easily ingested and compared, the following standards will be used: *i)* Time-Series Data: All time-series data, whether originating from the physical IoT network or generated by the Predictive Indoor Climate AI Service, will be accessible via REST APIs that deliver data in a standardised JSON format or as downloadable CSV files, *ii)* Simulation Raster Outputs: The quantitative, spatial outputs from the WDR and solar radiation simulations (e.g., heatmaps, rain load maps) will be provided in standard geospatial formats like GeoTIFF or NetCDF. This allows the data to be used for advanced quantitative analysis in external GIS software or scientific programming environments.
- **Metadata, Schema and Ontology Alignment:** This is the most critical layer for ensuring true interoperability. In relation to the metadata schema: all data assets and analytical events will be described by metadata records in the ARTEMIS KB. These records will strictly adhere to the standardised JSON schemas defined in the "Knowledge Base Integration" section. This ensures that every piece of information is consistently described and discoverable. Furthermore, in relation to ontological alignment and to ensure that data is not just syntactically but also semantically interoperable (i.e., machine-readable and unambiguous), key concepts will be mapped to established ontologies and controlled vocabularies. For instance: *i)* Heritage concepts (e.g., "building," "room," "tapestry") will be aligned with the CIDOC Conceptual Reference Model (CRM) where applicable, *ii)* Environmental parameters (e.g., "Air Temperature," "Relative Humidity") will be mapped to terms from a relevant scientific ontology like the Semantic Web for Earth and Environmental Terminology (SWEET). This alignment ensures that a query for "Air Temperature" will correctly find data from all different sensor types and simulations, creating a powerful and truly integrated knowledge system.



- **Data Access and Sharing:** The RHDT will implement a secure, role-based access control system to manage permissions (e.g., public read-only, researcher-level access, curator-level admin rights). All data will be transmitted upon request and remain stored securely at KIK-IRPA. While the specific details will be formalised in a comprehensive Data Management Plan (DMP), a tiered access model will be implemented. Full access to the complete, raw sensor and simulation data will be granted to all official ARTEMIS project partners and the formally designated representatives from the partner heritage institutions for the purposes of collaborative development, direct asset management and in-depth scientific analysis. Furthermore, for the purpose of end-user evaluation, participating heritage professionals will be granted temporary, but comprehensive, access to the relevant datasets through the RHDT platform for the duration of the testing period, governed by strict confidentiality agreements. For public dissemination materials such as video demonstrators and scientific publications will be made openly accessible. This framework ensures that while the project's methodologies and findings are shared widely, the raw, high-resolution data remains protected, with a strict policy prohibiting its redistribution outside of the authorised parties.

Available Metadata and Paradata: For the Japanese Tower pilot, a core activity will be the systematic creation and documentation of metadata and paradata, as much of this information does not yet exist in a structured format. The framework below outlines what is currently available and what will be produced during the project.

- **Administrative Metadata:** The Japanese Tower is a protected monument, property of the Belgian Federal State and is managed by the Belgian Buildings Agency. The collections within, including some historic furniture, are part of the Royal Museum of Art and History. All pilot activities, including non-invasive 3D scanning and the temporary placement of IoT sensors, are governed by a formal collaboration agreement between KIK-IRPA and these authorities. As all involved parties are public federal authorities, the protocols for data rights, access conditions and long-term preservation will be developed in full alignment with the European directive on open data.
- **Descriptive Metadata:** An extensive inventory of existing heritage documentation (scholarly articles, historical records, restoration reports) and photographic documentation (historical and recent images) is available as part of the collection status of the Japanese Tower and its interior elements within the collections of the Royal Museum of Art and History. A primary task of this pilot will be to create rich, new descriptive metadata by:
 - Spatially Annotating the 3D Model: Linking specific paragraphs from historical documents, archival photographs and analytical data directly to their corresponding locations on the 3D reconstruction of the tower.
 - Documenting the "As-Built" and "As-Is" States: Creating detailed descriptions for both the reconstructed 1905 state and the current state, including information on materials, decorative techniques and condition assessments.
 - Linking to Ontologies: Key entities (e.g., the tower itself, specific lacquer panels, material types) will be mapped to established ontologies, primarily

CIDOC CRM, to ensure semantic interoperability and facilitate machine-readability.

As the IoCT network is deployed, each sensor's placement will be documented with records including: room ID/code, floor, room name, sensor type, short description/use and its relationship to the building fabric. This information will be maintained in a structured format (e.g., CSV or a simple database) and mapped as annotations onto the RHDT, making the monitoring network fully discoverable and understandable within the 3D environment.

- **Structural Metadata:** Currently, no formal structural metadata exists. It will be created during the pilot to define the organisation and relationships between the project's heterogeneous datasets. The structure will be managed within the ARTEMIS KB, which will create a hierarchical and relational map linking the primary asset (the Japanese Tower) to all its associated components: its 3D models, the deployed IoT sensors, the historical source documents and every simulation run generated by the Predictive AI model.
- **Technical Metadata:** Systematic technical metadata is not yet available, but will be generated for all new digital assets created within the pilot, following the specifications outlined in the "Data Format Compatibility" section. This will include file formats, file sizes, creation dates and the specific software and versions used (e.g., Agisoft Metashape, Python libraries).
- **Paradata:** There is currently no existing paradata. However, paradata will be meticulously documented to capture the human processes and decisions behind the data creation during the project:
 - 3D Data Acquisition: As a new 3D model will be created, a detailed processing report (similar to an Agisoft Metashape report) will be generated. This will provide a complete record of the photogrammetric/laser scanning process, including acquisition parameters, calibration details, georeferencing specifics and mesh/texture processing settings.
 - IoCT Network Deployment: The rationale behind the placement of each sensor will be documented. This will capture the decision-making process involving conservators and heritage scientists, explaining why a specific location was chosen for monitoring.
 - Simulation and AI Model Development: This will be a critical paradata component. For each service, documentation will be created to record: *i)* Assumptions and Hypotheses: The scientific assumptions made in the model (e.g., material properties assumed for the exposed light simulation), *ii)* Decision-Making: The reasoning for choosing specific simulation parameters, *iii)* Processing Steps: The methodology used to train the Predictive Indoor Climate AI model, including the data cleaning, feature engineering and validation processes.
- **Data Quality Metadata:** Data quality will be documented through a composite of sources: *i)* The paradata reports from 3D model reconstruction will provide metrics on geometric accuracy (e.g., RMSE), *ii)* The manufacturer's technical specifications for all deployed IoCT sensors will document their accuracy and precision (e.g., $\pm 0.5^{\circ}\text{C}$

for temperature), *iii*) A key pilot activity will be to generate new data quality metadata by validating the simulation and AI models. This will involve quantitatively comparing the outputs of the predictive models against real-world data collected by the IoT sensors, providing a clear metric of their performance and reliability.

- **Usage History:** There is currently no documented usage history of the assets.

6.4.6 Pilot 14 - Data Collection and Data Types

This section outlines the methods and tools for data collection and specifies the types of data to be used for evaluating the pilot.

Data Types: For the pilot case a comprehensive set of environmental and material data will be collected from both inside and outside to provide a holistic view of the conditions affecting the Japanese Tower.

- **Sensor Data on the Indoor Environment:** air temperature & surface temperatures, relative humidity (RH), light levels (visual, infrared and ultraviolet), airborne dust particles, carbon dioxide (CO₂) concentrations
- **Sensor Data on the Outdoor Environment:** standard meteorological data from an on-site weather station, including temperature, relative humidity, rainfall, wind direction, wind speed and solar radiation
- **Periodic Diagnostic Data:** infrared (thermal) Imaging to identify thermal bridges, moisture anomalies and air leakage paths, hyperspectral Imaging to detect moisture presence and distribution and early signs of material decay or biological growth not visible to the naked eye

Data Collection Methods:

- **Tools and Technologies:** Data will be collected using a combination of commercial, off-the-shelf sensor devices and custom-built sensors designed to meet specific pilot requirements. Weather stations will be used for outdoor data, while specialised thermal and hyperspectral cameras will be used for periodic data capture.
- **Data Logging and Frequency:** The primary environmental data will be collected via automatic logging at intervals of 10 to 15 minutes, ensuring a high-resolution temporal dataset. The periodic diagnostic data (moisture checks, IR/hyperspectral imaging) will be collected manually once each season to capture seasonal variations and provide deeper diagnostic insights.
- **Data Format and Transmission:** Raw sensor data will be stored locally in simple, robust formats (.csv or .txt). For real-time integration, the data will be transmitted from the sensors to the RHDT platform using the MQTT protocol.

Expected Outcomes:

- **Primary Outcomes:** The main outcome will be a multi-layered, interactive application that functions as both a diagnostic tool and a storytelling platform. More specifically, the primary outcomes will be: *i*) An expert diagnostic tool: Allowing

conservator-restorers and researchers to visualise the reconstructed Japanese Tower, run historical environmental simulations and quantitatively analyse the impact of light and climate on its degradation, *ii*) A public-facing VR/AR experience: An immersive, narrative-driven application that will serve as a powerful, persuasive tool to justify restoration budgets, support fundraising efforts and visually communicate the urgency and importance of the conservation work. In addition, for the general public, it will make the inaccessible monument and its complex history engaging and understandable.

- **Secondary Outcomes:** The secondary outcomes are the following: *i*) A validated environmental simulation model: The pilot will produce a scientifically-grounded simulation model of the historical climate and light impact inside the Japanese Tower. Crucially, this model's accuracy will be calibrated and validated against the real-world data collected by the IoT sensor network, transforming it from a theoretical exercise into a reliable diagnostic tool. It will lay down the foundations for other environmental simulation models, *ii*) A foundational dataset for conservation planning: The combination of detailed 3D models, historical research and the validated simulation results will form a comprehensive evidence base. This will directly inform the creation of a future physical, climate change robust conservation-restoration strategy for the Japanese Tower's unique and fragile decorations, *iii*) A high-resolution environmental calibration dataset: A continuous, multi-variable time-series dataset will be collected from the on-site IoT sensors, providing a wealth of information on the Japanese Tower's current indoor/outdoor environmental interaction. While its primary use in this pilot is to calibrate the historical model, this dataset will also serve as an essential training resource for any future AI-based solutions aimed at developing predictive models or issuing warnings based on weather forecasts.

6.4.7 Pilot 14 - Evaluation and Validation Metrics

This section outlines the evaluation approach for the pilot, based on key performance indicators (KPIs) and other relevant success metrics.

Success Criteria: The following specific, measurable indicators will determine the pilot's success in achieving its objectives for monitoring, anomaly detection and Knowledge-Base integration. These criteria are designed to validate the entire workflow, from the reliability of the underlying data collection to the effectiveness of the advanced simulation and decision-support services. The metrics are aligned with the pilot's goals and project objectives:

- **Tier 1: Foundational System Performance:** These criteria ensure the underlying data infrastructure is robust and reliable: *i*) Sensor Data Reliability: At least 95% of expected readings per day from the active IoT network are successfully recorded, transmitted and stored in the relevant datastore, *ii*) System Stability: Sensors operate continuously with minimal dropouts or power interruptions, *iii*) KB Synchronisation: Metadata for all deployed sensors, completed simulations and generated forecasts are correctly registered in the ARTEMIS KB within an acceptable time frame (e.g., within 1 hour of the event), *iv*) Data Visualisation: Live and historical IoT sensor data



streams are successfully retrieved via the KB and visualised on the correct locations within the 3D models in both the 3D Viewer and the AR/VR Experience.

- **Tier 2: Advanced Service and Simulation Performance:** These new criteria evaluate the core innovations of Pilot 14.
 - Predictive Forecasting Accuracy
 - Description: The Predictive Indoor Climate AI Service must provide reliable and accurate forecasts that are trustworthy for proactive decision-making.
 - Metric: The AI model must predict interior temperature and relative humidity over a 24-hour forecast horizon with an accuracy that meets a predefined target (e.g., within $\pm 1^{\circ}\text{C}$ and $\pm 5\%$ RH), as validated against the real data collected by physical sensors.
 - Risk Assessment and Alerting
 - Description: The Risk Assessment Service must correctly translate all integrated data (real-time, forecasted and simulated) into accurate and timely risk alerts based on user-defined thresholds.
 - Metric: The system correctly identifies and logs risk events from all three data sources (real-time measurements, AI forecasts and climate simulation outputs) and demonstrates that alerts are triggered reproducibly for identical conditions.
 - Collaborative VR/AR Functionality
 - Description: The immersive environment must function as an effective tool for collaborative, multi-layered data analysis.
 - Metric: At least two users can successfully collaborate in a shared VR session, simultaneously visualising, annotating and discussing at least two different data layers (e.g., live IoT data overlaid with a WDR simulation heatmap) on a 3D model.
- **Tier 3: User Impact and Decision Support:** This tier measures the ultimate success of the pilot: its value to the end-users.
 - Decision Support Effectiveness
 - Description: The RHDT must empower heritage professionals to make more informed and proactive preservation decisions.
 - Metric: In post-pilot surveys and interviews, at least 70% of participating heritage professionals agree or strongly agree that the platform provides actionable insights and increases their confidence in making timely conservation decisions.

User Feedback: User feedback will be systematically gathered through a structured, mixed-methods approach during facilitated evaluation sessions. This approach combines hands-on testing with systematic data collection to elicit both quantitative metrics and rich qualitative insights relevant to the specific heritage context of each case study. The methods will include:

- **Facilitated, Scenario-Based Testing Sessions:** Small groups of users (1-3 participants), including heritage managers, conservators and other relevant professionals, will be guided through a series of realistic tasks. These tasks will



leverage the RHDT platform and the collaborative VR/AR environment to simulate real-world decision-making scenarios, such as inspecting temperature and humidity trends, or comparing the impacts of different climate change scenarios.

- **Think-Aloud Protocols and Real-time Observation:** During these sessions, facilitators will encourage users to verbalise their thought processes and will systematically record observations using standardised checklists. This will capture user hesitations, errors, points requiring assistance, successful task completions during the evaluation tasks and overall interaction flow.
- **Post-session Surveys:** Following the hands-on tasks, participants will complete short, structured questionnaires (10-12 items) using 5-point Likert scales. These surveys will assess key aspects like data clarity, ease of interpretation of complex visualisations (including real-time IoT data, simulation results and anomaly logs), perceived accuracy and trustworthiness of the predictive models and overall system usability within their professional context.
- **Expert Interviews/Focus Groups:** Brief debrief sessions (10-15 minutes) or focused discussions will be conducted with conservation professionals to capture in-depth qualitative feedback. This will explore the usefulness of the RHDT workflow, its potential integration with existing monitoring practices, the perceived reliability of alerts and forecasts and its overall value for professional decision-making and strategic planning.
- **Description of the Collected Feedback:** The feedback collected will aim to assess the RHDT's performance across multiple dimensions, relevant to the unique challenges of each case study:
- **Data Clarity and Usability:** The ease with which users can understand and interpret the visualised IoT sensor data, anomaly logs and the outputs from the Predictive Indoor Climate AI services, whether on a 3D interface or in a VR/AR environment.
- **Effectiveness of Predictive Tools:** Users' perceived accuracy, trustworthiness and actionable utility of the AI-driven indoor climate forecasts and the simulation services. This includes assessing whether users can confidently act upon the provided predictions.
- **Impact on Professional Workflow:** How the RHDT and its advanced tools (simulations, forecasts) could potentially alter or improve their existing professional practices for monitoring, risk assessment and conservation planning.
- **Collaborative VR/AR Experience:** For use cases involving the AR/VR environment, the effectiveness of the immersive environment for collaborative analysis and communication, enabling shared understanding of complex environmental data and simulated impacts.
- **Use of Feedback:** The collected feedback will be systematically analysed to:
 - **Iteratively Refine the RHDT Platform:** Identify specific areas for improvement in the user interface, data visualisation, alert mechanisms and the integration of the simulation outputs.
 - **Develop Best-Practice Guidelines:** Compile insights from heritage professionals into evidence-based recommendations for implementing similar digital twin technologies in heritage conservation.



- Validate Against Success Criteria: Quantitatively measure user engagement, perceived value and the platform's contribution to informed decision-making, directly aligning with the pilot's defined success criteria.

Results will be compiled into pass/fail matrices aligned with success criteria and summary reports for WP12.

Performance Metrics: The key technical parameters that will be measured to quantitatively evaluate the performance, reliability and responsiveness of the pilot's end-to-end infrastructure. These metrics cover the entire data pipeline, from the physical IoT sensors to the advanced simulation services and the final user-facing visualisation environments.

- IoT Network and Data Ingestion Performance: These metrics evaluate the performance of the physical sensor network and the data ingestion pipeline under real operational conditions.
 - System Uptime: Percentage of continuous operational availability of all deployed IoT sensor nodes during the active evaluation phase. Target: $\geq 90\%$
 - Data Completeness: Percentage of days with full 24-hour datasets and valid daily averages for each key monitored location. Target: $\geq 90\%$
 - Data Transmission Latency: The average delay between a sensor reading being taken and its successful registration and availability in the datastore. Target: < 5 seconds
- Simulation and Analytical Services Performance (Backend): These metrics evaluate the computational performance and responsiveness of the developed backend services.
 - Predictive Forecast Generation Time: The time required for the Predictive Indoor Climate AI Service to generate a full 24-hour forecast after receiving the external weather data. Target: < 5 minutes
 - Risk Assessment Latency: The time taken by the Risk Assessment Service to process incoming data (live or forecasted) and return a risk event status. Target: < 1 second
- RHDT Platform and visualisation Performance (Front-End): These metrics evaluate the performance of the user-facing applications, including both Web-based and immersive VR environments.
 - Scene and Data Loading Time: The average time required to open an RHDT scene and display the initial data overlays (e.g., live temperature readings). Target: < 10 seconds
 - VR Rendering Performance: The average frame rate (FPS) when navigating and interacting with complex 3D scenes (including multiple data layers) in the VR environment using Meta Quest 3 headsets. Target: ≥ 72 FPS (for a smooth, immersive experience)
 - User Interaction Responsiveness: The measured latency between a user input (e.g., selecting an anomaly, rotating the model, toggling a simulation layer) and the corresponding on-screen visual response. Target: < 0.5 seconds.



- Visual Clarity and Readability: Percentage of participants who rate the readability of colour-coded climate data and anomaly labels as “clear” or “very clear” in post-session surveys. Target: ≥ 90 %
- Overall System Communication Performance: This metric evaluates the efficiency of the communication “glue” that connects all parts of the ARTEMIS architecture.
 - API Response Time: The average time for any cloud-based REST API call made from the front-end (e.g., to fetch IoT data, trigger a risk assessment, or retrieve a simulation result) to return a processed data payload. Target: < 2 seconds per call

6.4.8 Pilot 14 - Essential Features Specification Checklist

The following checklist serves as a rapid reference tool for assessing the pilot’s alignment with the ARTEMIS objectives and features described in the DoA.

Property category	Key Functionality Points
Methodology & Simulation	Converts theoretical knowledge into simulation services
	Simulates the effects of natural processes and climate events on heritage assets.
	Perform decision-making
Diverse Case Studies	Addresses tangible aspects of CH
Real-World Demonstration and test	Provides a real-world example of ARTEMIS service usage
AR/VR Integration	Employs AR/VR techniques for immersive visualisation
	Supports client-side online 3D visualisation
	Supports layered information
User Interaction & Visual Feedback	Represents the effects of human interventions visually
	Represents the effects of human interventions numerically
Technical & Data Capabilities	Visualise large-scale datasets on low-end devices
	Stream output to browser-based clients
	Real-time IOT data collection and analysis

Table 6.20 - Pilot 14 - Essential Features Specification Checklist